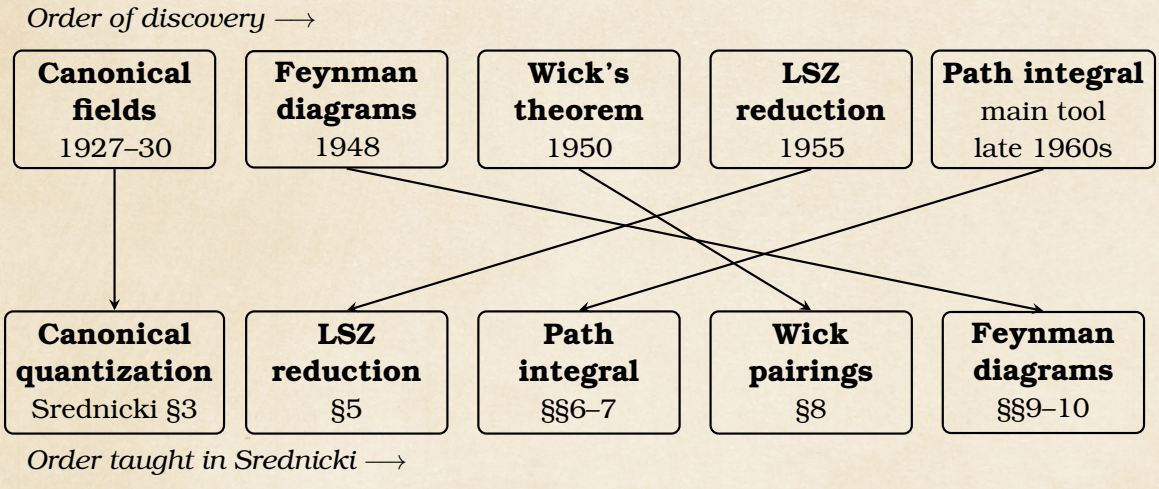
10.1 How QFT Got Here: A Short History

QFT was built in close to the reverse of the order we teach it. Fields were quantized first (1927–30), Feynman diagrams arrived in 1948, Wick's theorem in 1950 and LSZ in 1955. The path integral became the everyday working tool only in the late 1960s.

Srednicki runs the story backwards on purpose: LSZ says what to compute, the path integral computes it, and Feynman rules fall out of Gaussian integrals. The history shows why each step was needed, and which problem each one solved.



Only canonical quantization keeps its place. LSZ, the last of the operator results, moves up to second, and Feynman diagrams, the first computational tool, move to last.

Timeline at a glance

Year	Who	What
1926	Born, Heisenberg, Jordan	Jordan quantizes a free field and recovers Einstein's 1909 fluctuation formula
1927	Dirac	Quantized radiation field; spontaneous emission from first principles
1927–28	Jordan, Klein; Jordan, Wigner	Matter fields quantized ("second quantization"); anticommutators for fermions
1929–30	Heisenberg, Pauli	The canonical recipe for fields: $\mathcal{L}$, $\pi = \partial\mathcal{L}/\partial\dot{\varphi}$, equal-time commutators
1933	Dirac	The amplitude "corresponds to" $e^{iS/\hbar}$
1934	Pauli, Weisskopf	Charged scalar field: antiparticles without a Dirac sea
1939–40	Wigner; Pauli	Unitary irreps of the Poincaré group; spin–statistics (free fields)
1942	Feynman	Path integral (PhD thesis; published 1948)
1947	Lamb, Retherford; Kusch, Foley; Bethe	Lamb shift and electron $g-2$ measured; first renormalized estimate
1946–49	Tomonaga, Schwinger	Covariant, renormalized QED in operator form
1948	Feynman	Diagrams first shown, at the Pocono conference
1949	Dyson	Feynman's rules derived from operator QED; renormalizability argued
1949	Feynman	Propagator and diagrams in print (Phys. Rev. **76**)
1950	Wick	Wick's theorem
1955	Lehmann, Symanzik, Zimmermann	LSZ reduction formula
1967	Faddeev, Popov	Ghosts from gauge-fixing the path integral
1971–72	't Hooft, Veltman	Yang–Mills theory renormalized
1974	Wilson	Lattice gauge theory

Why fields (1926–1940)

Fields became unavoidable once relativity met quantum mechanics: particle number is not conserved, and causality needs antiparticles. It took about a decade to see this clearly.

- **Radiation first.** Treating each mode of the electromagnetic field as

an oscillator predates quantum mechanics. Jordan quantized a free field in 1926, and Dirac's 1927 paper derived spontaneous emission from photon creation and annihilation operators. Dirac's paper is usually called the birth of quantum electrodynamics.

- **Then matter.** Jordan and Klein (1927) and Jordan and Wigner (1928) quantized the Schrödinger wave itself, with anticommutators for fermions. Heisenberg and Pauli (1929–30) gave the general canonical recipe that our first lectures still follow.
- **Then antiparticles.** The Dirac equation (1928) led to hole theory and an infinite sea of filled negative-energy states. Pauli and Weisskopf (1934) quantized the charged Klein–Gordon field and got particles and antiparticles, both with positive energy, and no sea. Fermi's β-decay theory (1934) created an electron and a neutrino at a point.

Wigner (1939) classified the unitary irreducible representations of the Poincaré group, the basis of the modern definition of a particle. Pauli (1940) proved the spin–statistics theorem for free fields. Weinberg later recast the whole argument: QFT is what a Lorentz-invariant, unitary S-matrix with cluster decomposition requires (his Vol. 1, chs. 2–5).

Infinities, the 1947–49 breakthrough, and Feynman's diagrams

From 1930 on (Waller, Oppenheimer), QED calculations beyond lowest order diverged, and the 1947 experiments forced a fix. Many physicists had concluded that QFT was broken. Heisenberg's S-matrix program (1943), developed independently of Wheeler's 1937 S-matrix, responded by talking only about observable scattering amplitudes, an idea LSZ later revived.

In 1947 Lamb and Retherford measured the $2S$–$2P$ splitting in hydrogen, and Kusch and Foley found the electron's anomalous magnetic moment. Bethe's quick estimate of the Lamb shift, using mass renormalization, gave about 1040 MHz against roughly 1000 MHz measured. Four people then built covariant, renormalized QED:

- **Tomonaga** (wartime Tokyo, in English 1946): a covariant interaction picture on spacelike surfaces.
- **Schwinger** (1947–49): covariant operator renormalization, and $a = \alpha/2\pi$ for the electron's anomalous moment.
- **Feynman** (1947–49): an intuitive space-time scheme, checked against Schwinger's numbers.
- **Dyson** (1949): showed the three equivalent and argued renormalizability to all orders. The complete proof (BPHZ) came in 1957–69.

The diagrams. Feynman built on his 1942 path-integral thesis and on Wheeler's suggestion, around 1940, that a positron is an electron moving backward in time; Stueckelberg had the same idea independently in 1941–42. Feynman first showed diagrams at the Pocono conference (30 March–2 April 1948), where Bohr objected that electron trajectories violate the uncertainty principle. Dyson derived Feynman's rules from operator QED in a paper published in February 1949, before Feynman's own papers that September. Dyson's version made the diagrams teachable.

The propagator. The Klein–Gordon operator has several Green's functions. Feynman chose the one that carries positive frequencies forward in time and negative frequencies backward; that is his backward-moving positron, written as a boundary condition. In Srednicki's conventions,

$$\Delta(x-y) = \int \frac{d^4k}{(2\pi)^4}\,\frac{e^{ik(x-y)}}{k^2+m^2-i\epsilon}, \qquad \langle 0|\,T\varphi(x)\varphi(y)\,|0\rangle = \frac{1}{i}\,\Delta(x-y). \tag{10.1.1}$$

Stueckelberg and Rivier introduced an equivalent causal propagator in 1948, and Feynman's 1949 paper credits Stueckelberg, but his work, mostly in French, was little noticed at the time. Dyson identified the propagator with the vacuum expectation value of a time-ordered product. Why the vacuum at both ends selects this Green's function is the subject of §10.8.

Wick's theorem (1950)

Wick's theorem turned Dyson's expansion into a mechanical rule: every time-ordered product of free fields is a sum over ways of pairing them into propagators. Dyson's series gave each order of the S-matrix as such a product, but people were reducing them to matrix elements case by case. G.-C. Wick proved the general identity in "The Evaluation of the Collision Matrix" (Phys. Rev. **80**, 268, 1950). Each fully paired term is one Feynman diagram, fermion signs included.

The combinatorics is older. Isserlis (1918) proved the same pairing rule for moments of Gaussian random variables, and a free field is a Gaussian integral. That is why Srednicki §8 needs no operator version: expanding $Z_0(J)$ gives the pairings directly, for example

$$\langle 0|\,T\varphi(x_1)\varphi(x_2)\varphi(x_3)\varphi(x_4)\,|0\rangle = \frac{1}{i^2}\big[\Delta_{12}\Delta_{34}+\Delta_{13}\Delta_{24}+\Delta_{14}\Delta_{23}\big], \tag{10.1.2}$$

with $\Delta_{ij}\equiv\Delta(x_i-x_j)$. Peskin & Schroeder §4.3 takes the operator route instead. The Wick rotation to Euclidean time is a separate result of Wick's, from 1954.

LSZ (1955)

LSZ defines scattering using only the interacting fields and physical quantities, with no appeal to free fields or the interaction picture. Lehmann, Symanzik and Zimmermann, whom Pauli nicknamed the "Feldverein", published it in Nuovo Cimento **1**, 205 (1955). By 1955 Dyson's derivation looked shaky:

- It leaned on the interaction picture and adiabatic switching (Gell-Mann and Low, 1951), which assume the interacting vacuum and particles are simply related to free ones.
- Renormalization undercut that. The physical mass differs from the bare mass, and the Källén–Lehmann spectral representation (1952–54) showed that the field's normalization Z lies between 0 and 1.
- Haag (1955) argued, and Hall and Wightman (1957) proved, that an interaction picture cannot exist in a nontrivial relativistic theory.

The key assumption of LSZ is the asymptotic condition: as $t \to \mp\infty$, φ approaches $Z^{1/2}\varphi_{\text{in/out}}$. The limit holds only weakly, in matrix elements between normalizable states (§10.6); as an operator statement it would make the theory free. (Z is the residue of the propagator's one-particle pole, Peskin and Schroeder's Z, so $\langle k|\varphi(0)|0\rangle = \sqrt{Z}$.) The reduction formula follows (Srednicki §5). Below is his (5.15), which holds for the field rescaled by $Z^{-1/2}$ so that $\langle k|\varphi(0)|0\rangle = 1$, as he requires after (5.18); for the unrescaled φ each leg carries one more factor $Z^{-1/2}$:

$$\langle f|i\rangle = i^{\,n+n'} \int d^4x_1\, e^{ik_1x_1}(-\partial_1^2 + m^2)\cdots \int d^4x'_1\, e^{-ik'_1x'_1}(-\partial_{1'}^2 + m^2)\cdots \\ \times \langle 0|\, T\varphi(x_1)\cdots\varphi(x'_1)\cdots|0\rangle. \quad (10.1.3)$$

In words: an S-matrix element is the on-shell residue of a time-ordered correlator, with each external leg amputated. Haag (1958) and Ruelle (1962) later constructed scattering states rigorously for theories with a mass gap, and Hepp (1965) derived the LSZ asymptotic condition within that framework.

The path integral

The path integral was invented in the 1940s, sat mostly unused for twenty years, and became the default language of QFT only once non-abelian gauge theories needed it.

- **Origins.** Dirac (1933) noticed that the transition amplitude "corresponds to" $\exp\!\big(i\int L\,dt/\hbar\big)$; he wanted a Lagrangian, and so relativity-friendly, quantum mechanics. Feynman made

"corresponds to" into "is proportional to" by summing over all paths, in his 1942 thesis. He needed it because Wheeler–Feynman electrodynamics has an action but no Hamiltonian.

- **Two quiet decades.** Feynman applied it to QED in 1950, and Schwinger's generating functional $Z[J]$ (1951) brought functional methods into operator language. Still, most calculations through the 1960s used operators.
- **Why it became indispensable.** Feynman (1963) found that one-loop Yang–Mills and gravity amplitudes break unitarity unless extra "ghost" loops are added. Faddeev and Popov (1967) derived the ghosts in a few lines by gauge-fixing the path integral. 't Hooft and Veltman then renormalized Yang–Mills theory (1971–72), and path-integral methods spread widely in the early 1970s.
- **The nonperturbative era.** Euclidean field theory, Wilson's renormalization group and lattice gauge theory (1971–74), instantons (1975–76), and anomalies as the Jacobian of the fermion measure (Fujikawa, 1979) all live naturally in the path integral.

Its limits matter too. The Minkowski "measure" is not mathematically defined, and rigorous constructions of interacting theories exist only in 2 and 3 spacetime dimensions. The operator picture is still what supplies the Hilbert space, unitarity, the spectrum and LSZ itself, which is one reason this course does canonical quantization first.

Further reading

- S. S. Schweber, *QED and the Men Who Made It* (Princeton University Press, 1994): the 1947–49 story in full, with the physics.
- S. Weinberg, *The Quantum Theory of Fields*, Vol. 1, ch. 1: a physicist's history of the subject's early decades.
- D. Kaiser, *Drawing Theories Apart* (University of Chicago Press, 2005): how Feynman diagrams spread after 1949.
- A. Pais, *Inward Bound* (Oxford University Press, 1986): the wider history of particle physics.

Lecture asides

Two short asides for the lectures.

Aside A, for LSZ (LecNote4, Srednicki §5): why LSZ, and why 1955?

Feynman diagrams worked by 1949, but their derivation assumed the interacting vacuum and particles are just the free ones, slowly switched on. Renormalization showed that cannot be right: the physical mass is not the bare mass, and φ creates a one-particle state with strength $\sqrt{Z}$, not 1 (Z is the residue of the propagator's pole, as in Peskin and Schroeder; LecNote4 p. 8 calls the strength itself Z). Lehmann, Symanzik and Zimmermann asked what follows from the interacting field alone. Their answer: far in the past and future, φ behaves like $\sqrt{Z}$ times a free field, but only inside matrix elements. That is enough to turn any time-ordered correlator into an S-matrix element. Shifting and rescaling φ so that $\langle 0|\varphi(0)|0\rangle = 0$ and $\langle k|\varphi(0)|0\rangle = 1$ is this normalization.

Aside B, for the path integral (LecNote5–8, Srednicki §§6–9): where this formalism came from

Dirac (1933) noticed that the quantum amplitude corresponds to $e^{iS/\hbar}$. Feynman, a Princeton graduate student, turned that into a sum over all paths in his 1942 thesis. For twenty years most physicists computed with operators instead. The path integral became essential in 1967, when Faddeev and Popov used it to quantize Yang–Mills theory. It is now the default language for gauge theories, the renormalization group and the lattice. The pairings in §8 are Wick's theorem (1950); in the path integral they are just Gaussian moments, a rule statisticians had in 1918.